



MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

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Sl.No.

SESSIONAL RECORD

DEPT.: COMPUTER SCIENCE
ENGG

NAME: RUTVIK AVINASH BARBHA

REG.NO.: 225805222

SEMESTER: VII

SECTION: CSE-CORE-A

ROLL NO.:

SUBJECT: CLOUD COMPUTING

DATE OF TESTS: 13/10/2025

MAX MARKS: 15/20

MARKS OF: I TEST

II TEST

28/30

MARKS FOR VALUATION - I TEST

MARKS FOR VALUATION - II TEST

Q. No.	FOR THE TEACHER TO AWARD MARKS					TOTAL MARKS	Q. No.	FOR THE TEACHER TO AWARD MARKS					TOTAL MARKS
	a	b	c	d	e			a	b	c	d	e	
1							1						
2							2						
3							3						
4							4						
5							5						
6							6						
TOTAL IN FIGURES							TOTAL IN FIGURES						

IN WORDS _____

IN WORDS _____

Ravi Barbhui

Signature of the Candidate

8-1

Signature of the Teacher

Topic: MCQs (Multiple Choice Questions)

- Q 1. option c) Elasticity
- Q 2. option b) Grid computing
- Q 3. option b) Live Migration
- Q 4. option b) Virtualization
- Q 5. option a) web service description language
- Q 6. option b) Elastic public cloud with auto-scaling group
- Q 7. option d) ^{elastic scaling} ~~security & regulatory compliance~~
- Q 8. option E) ^{Reservation phase} ~~iterative pre-copy reservation phase~~
- Q 9. option d) Each client request includes authentication
- Q 10. option c) Type-2 hypervisors on developer machine

Answer key:-

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	B	B	B	A	B	D	C	D	C

B

Q 11 A) Healthcare provider scenario:-

A11 a) Elasticity: Allows the provider to automatically scale telemedicine services and record access systems during healthcare emergencies or seasonal peaks (like flu season) and scale down during quiet periods ensuring cost efficiency

Resource Pooling: Enables access to massive computational power on demand for data intensive tasks like medical imaging analysis or genomic research which would be too expensive to host in house

Pay-per-use: The provider avoids huge upfront investment in IT infrastructure and instead pays only for the compute and storage resources they actually consume making budgeting and predictable and operations more affordable

A11 b) SaaS, PaaS, IaaS in Healthcare:-

Low / Software as a Service

Building and operating "infrastructure" services - setting a custom built hardware layer (IaaS) and/or setting up the servers for PaaS. These are capital - IT team manages the IT and software.

Low / Platform as a Service

Building a platform to build any sort of custom applications without worrying the underlying infrastructure.

Example: Developing a new Data portal app using AWS's like people and Engine.

Low / Software as a Service

Submitting to ready made application.

Example: using a cloud based medical billing software or a pre-built, third party ERP system like SAP.

(PHI) on a secure private cloud to maintain control and meet HIPAA compliance. Meanwhile less sensitive applications like the public facing website or a telemedicine video conferencing service, can run on a scalable cost-efficient public cloud.

All d) Hybrid cloud scenario

A hospital can host its core EHR database, containing sensitive patient records, on an on-premise private cloud, with strict access controls. A patient facing mobile App used for scheduling appointments and conducting video calls with doctors (example using Microsoft Azure as a platform) The App securely access necessary data from the private cloud using an encrypted API. This improves security by isolating the core, ~~services of the platform~~ by data while enhancing

Instead of setting up individual physical computers, an administrator can use the pre-configured VM template. They can then automate the provisioning process to rapidly deploy dozens of identical, ready to use environments for all students within minutes ensuring everyone has immediate access to the required tools

d) Migration For Availability

If a physical server hosting several critical VMs needs a reboot for a security update, the administrator can use live migrations to move those running VMs to another host in cluster. This allows the maintenance to be performed without any service downtime, ensuring the students experiments and application remain available

11B)e.

Balanced strategy

11B E) Implement a policy based automation strategy. Use VM template for rapid provisioning, allowing students to

to deploy their own environments via a self service portal based on predefined rules. Enable automated load balancing to automatically migrate VMs between hosts to prevent performance bottlenecks and optimize resource use without manual intervention.

Q12 A. Multinational company computing models.

12A. 9) Parallel computing for genome sequencing :-

Genome alignment is a massive comparison task. Parallel computing divides the large genome sequence into smaller segments and assigns each segment to a different processor core. All cores work simultaneously (in parallel) to perform the comparisons, and the results are combined at the end. This drastically reduces the total computation time.

Q 12A.6) Cluster computing for climate modelling
cluster computing is often more suitable than a single super computer because it is more scalable and cost-effective. A cluster is built for many commodity servers, making it cheaper to build and easier to expand by simply adding more nodes as climate models grow more complex. It offers better fault tolerance, as failure of one node does not bring down the entire system.

12 c) Distributed computing vs centralized computing

1. Centralized: A single, powerful mainframe computer processes all transactions. All branches and ATMs connect to this central point. It is easier to manage but represents a single point of failure.
2. Distributed: Processing is spread across multiple data centers in different regions. This provides higher availability and lower latency for customers but it is more complex to keep data consistent across all locations.

d) challenges in grid computing for drug discovery :-

A major challenge is data security and intellectual property protection. Drug discovery involves highly sensitive and proprietary data. When using a grid that pools resources from different organizations, ensuring this confidential data not exposed or compromised while being processed on third party machines is a significant technical and policy challenge.

e) Comparison of computing models :-

Parallel: Tightly coupled; resources are ~~sharing~~ shared within a single machine

Cluster: Tightly coupled; resources (homogeneous nodes) are shared over a high speed local network

Distributed: Loosely coupled; resources (heterogeneous nodes) are shared over a general purpose network.

Grid: Very loosely coupled. a form of distributed computing sharing heterogeneous resources across different administrative domains.

Q 126) Online shopping Architecture

a) Impact of Application Layer Failure:-

A failure in the business logic layer would be catastrophic. It would mean customers cannot process orders, use the recommendation engine, or make payments. This leads to an immediate loss of sales, severe customer frustration and damage to company's reputation.

b) Message Bus for Heavy sales:-

A message bus decoupled services. During a sale, the order service can place thousands of orders per minute onto the bus as messages. The payment and shipping services can then process these messages from the bus at their own maximum speed. This acts as a shock absorber, preventing the downstream services¹¹ from being overwhelmed.

c) Publish-subscribe for promotions!

The marketing system can publish a single "promotional offer" topic. Multiple other services can subscribe to this topic. The system automatically delivers the message to all subscribers, who then send the offer to users through their respective channels.

d) Payments Queue based vs. Publish-subscribe!

Queue based messaging is better for payments.

Reason: A payment transaction must be processed exactly once. In a queue, a message is delivered to ~~today~~ only one customer, even if many are listening. This ensures a customer isn't charged multiple times. Publish-subscribe would deliver the payment message to every subscriber leading to incorrect repeated charges.

e) Scaling the application Layer: PaaS vs IaaS :

PaaS (Platform as a Service) is better for quickly scaling

Reason :- PaaS platforms have built in, managed auto-scaling. A developer can simply deploy their code and the platform automatically handles the provisioning and configuration of additional resource instances in response to traffic. With IaaS, the team would have to manually configure the virtual machines, load balancers and scaling policies, which is significantly slower and complex.

FOR ROUGH WORK ONLY

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